

# EEG Projects

BioRTC 2026 · Yobe State University

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# Project 1: Seizure Detection using Deep Learning

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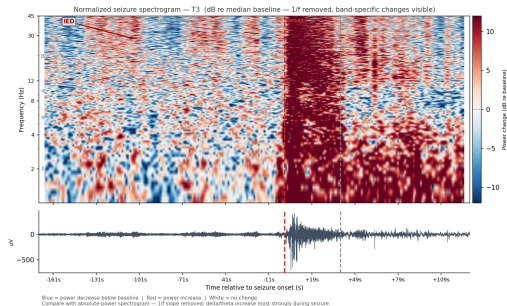
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## Dataset: TUSZ v2.0.6

3 000+ patients, expert seizure labels,

<5% of windows are seizures, 95% accuracy = detect nothing



Normalized spectrogram (dB re baseline), channel T3. Red = power increase; blue = decrease.

White vertical lines = seizure onset/offset. Post-ictal IEDs visible as bright vertical streaks.

# Project 1: What You Will Do

## Step 1 - Offline detection

- Preprocess TUSZ recordings
- Segment into 4 s windows, extract band-power features
- Train an **MLP** on features
- Train a **1D-CNN** on the raw signal
- Compare sensitivity / specificity / F1



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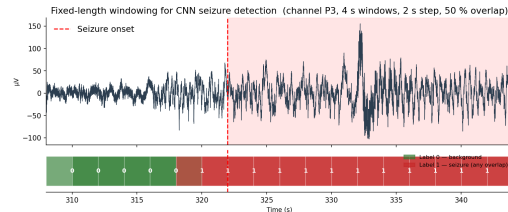
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4 s windows, 50% overlap. Red = seizure label. Note: seizure windows are a small fraction of the total.

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**Track:** CS / data science    **Group:** 2-4 students

**Notebooks:** EEG\_01 + EEG\_02

## Step 3 - Transfer to Yobe

Run on SUBJECT1.edf. No labels. Compare against the clinical report.

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## Project 2: Source Reconstruction

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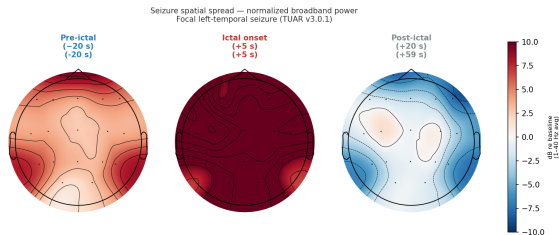
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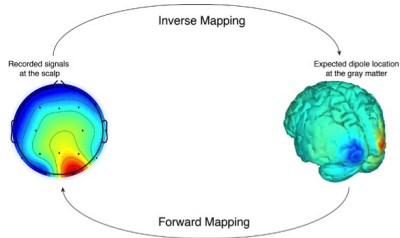
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Pre-ictal: resting baseline. Ictal: broadband power increase.

Post-ictal: power reduction. The scalp maps show *what* changes - but not *where in the brain* it originates.

# Project 2: The Inverse Problem

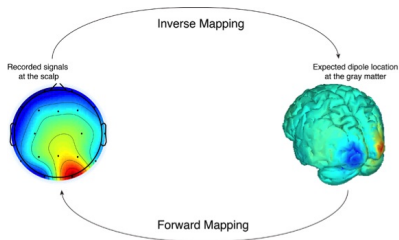


Forward problem (left): one dipole gives one unique scalp map. Inverse problem (right):

the same scalp map is consistent with many possible source configurations.



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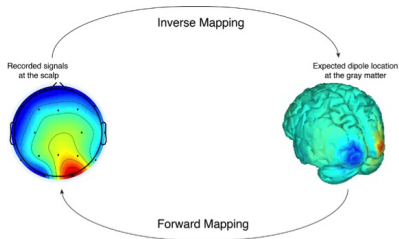
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## Three methods, one seizure

- **dSPM** - noise-normalized minimum norm; good spatial coverage
- **sLORETA** - zero localization error for point sources
- **LCMV beamformer** - suppresses activity from other locations

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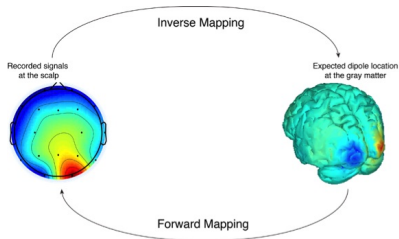
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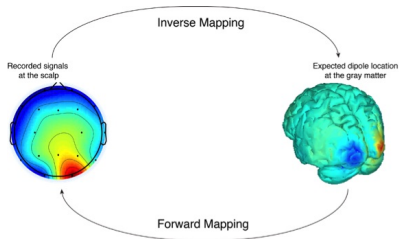
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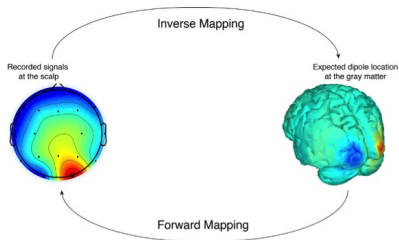
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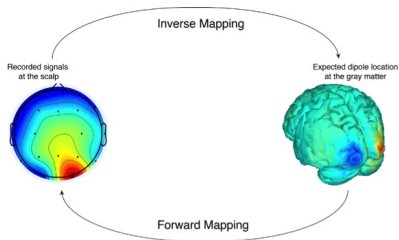
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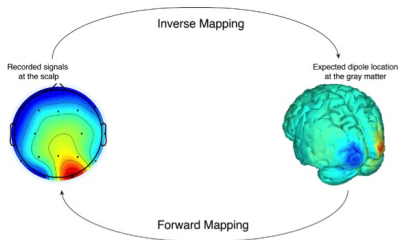
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## Project 2: What you will use

### Dataset: TUSZ v2.0.6

Same recordings as Project 1.

Standard **fsaverage** head model (no individual MRI needed).



|           | Project 1 - DL Seizure Detection   | Project 2 - Source Reconstruction |
|-----------|------------------------------------|-----------------------------------|
| Track     | CS / data science                  | Computational neuroscience        |
| Dataset   | TUSZ v2.0.6                        | TUSZ v2.0.6                       |
| Methods   | MLP, 1D-CNN                        | dSPM, sLORETA, LCMV               |
| Challenge | Class imbalance, online evaluation | Ill-posed inverse problem         |
| Extension | Transfer to Yobe EEG               | Localize in Yobe EEG              |
| Notebooks | EEG_01 + EEG_02                    | EEG_01 + EEG_03                   |

Both projects use the **same TUSZ recordings** and the same MNE-Python preprocessing pipeline.

Both end with an **open question**: does your method transfer to a real hospital recording from Yobe with no ground truth?